MACRO L3 - 2014-2015

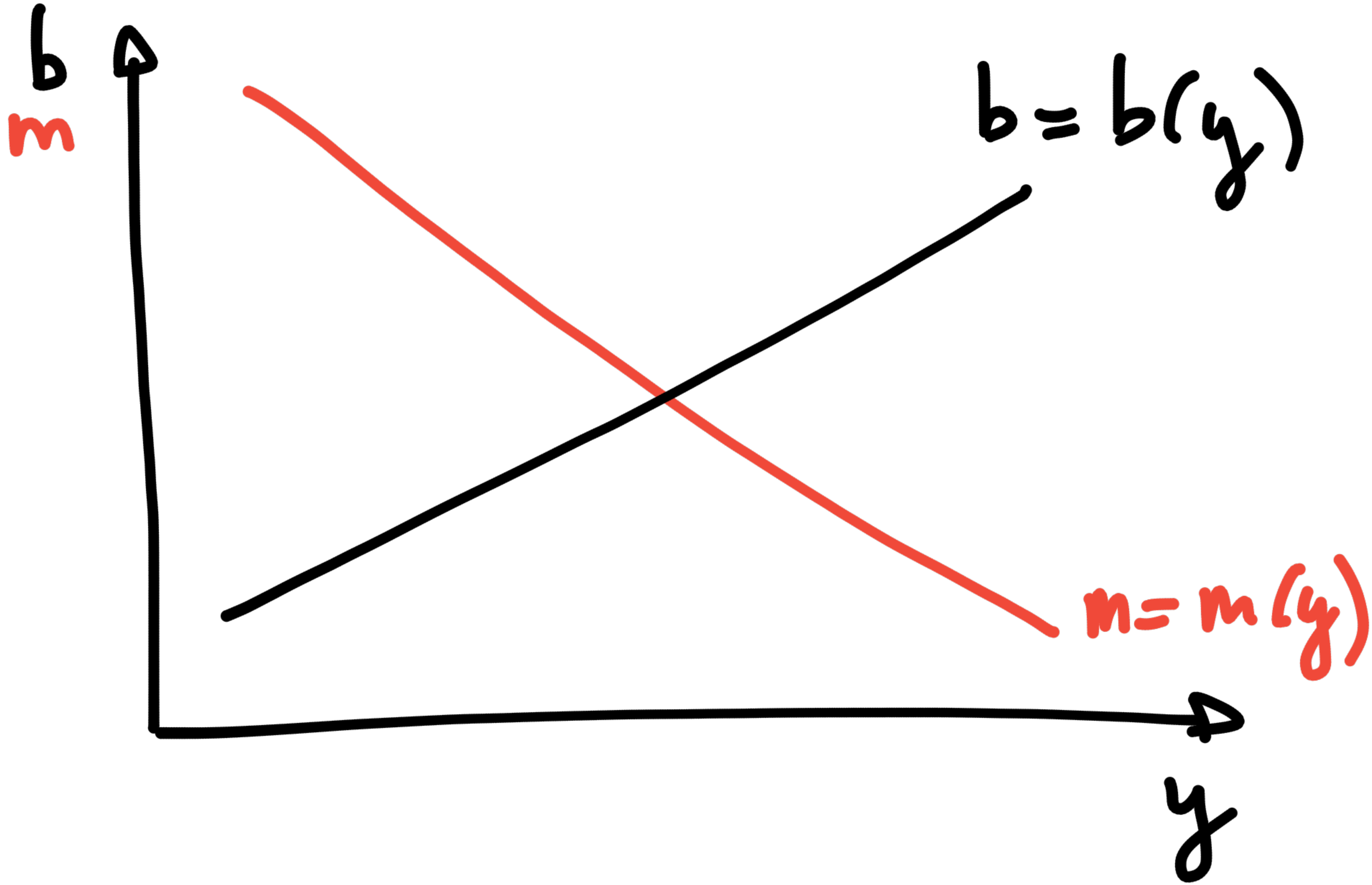
Lecture 5

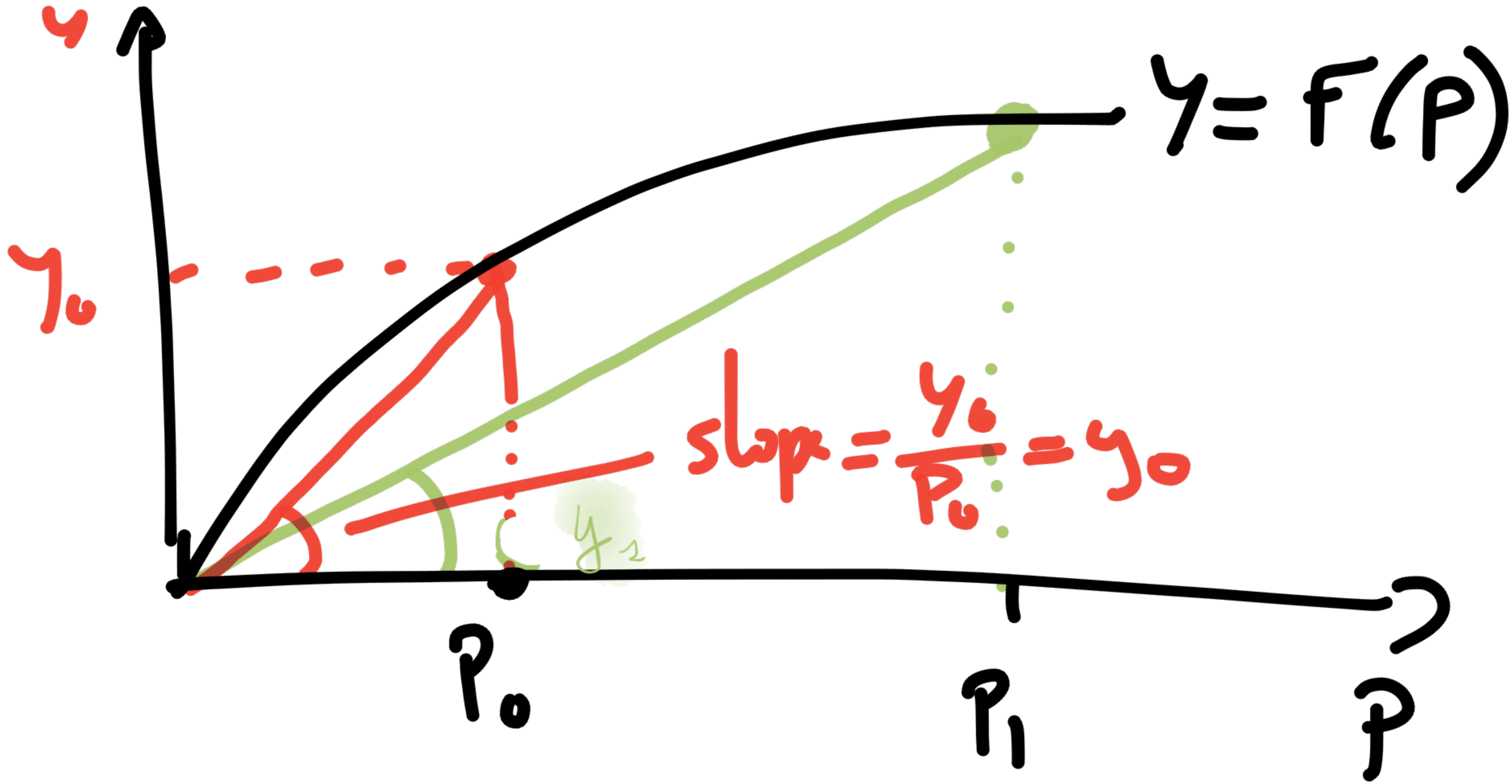
F. PORTIER

These notes are those I wrote during class. They are posted "for the record".

Session 2

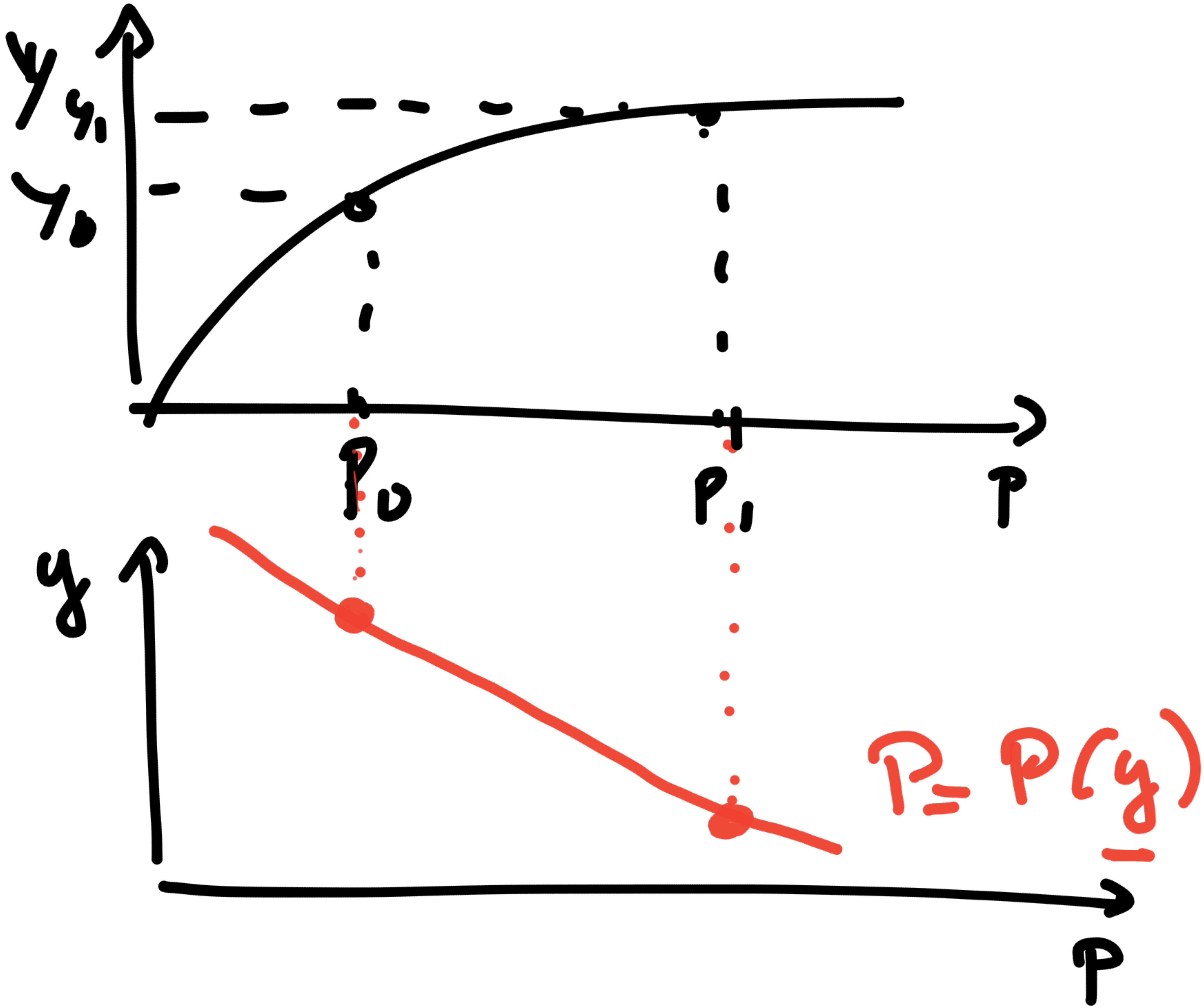
Nov 4, 2014


b
m
b = b(y)
m = m(y)
y



$$\frac{Y_0}{P_0} = y_0 = \text{income per capita}$$

$y_1 < y_0$



y
y_1
y_0
p_0
p_1
p
y
$p = p(y)$
p

Steady state : it is a situation in which population stays constant

\# of birth : $b \times \underline{P}$

\# of death : $m \times P$

Remark : if P is a function of time

2 choices

→ time is discrete : $t = 1, 2, 3, \ldots$

→ ——— continuous $t \in \mathbb{R}^+$

→ P_t, P_{t+1}, P_{t+2}

→ $P(t)$

change in P : $P_t - P_{t-1}$

$$\frac{P(t) - P(t-\partial t)}{\partial t}$$

$\partial t \to 0$ ↳ $\dfrac{dP(t)}{\partial t} = \dot{P}(t)$

$$\left(\Leftrightarrow P'(t)\right) = \dot{P}$$

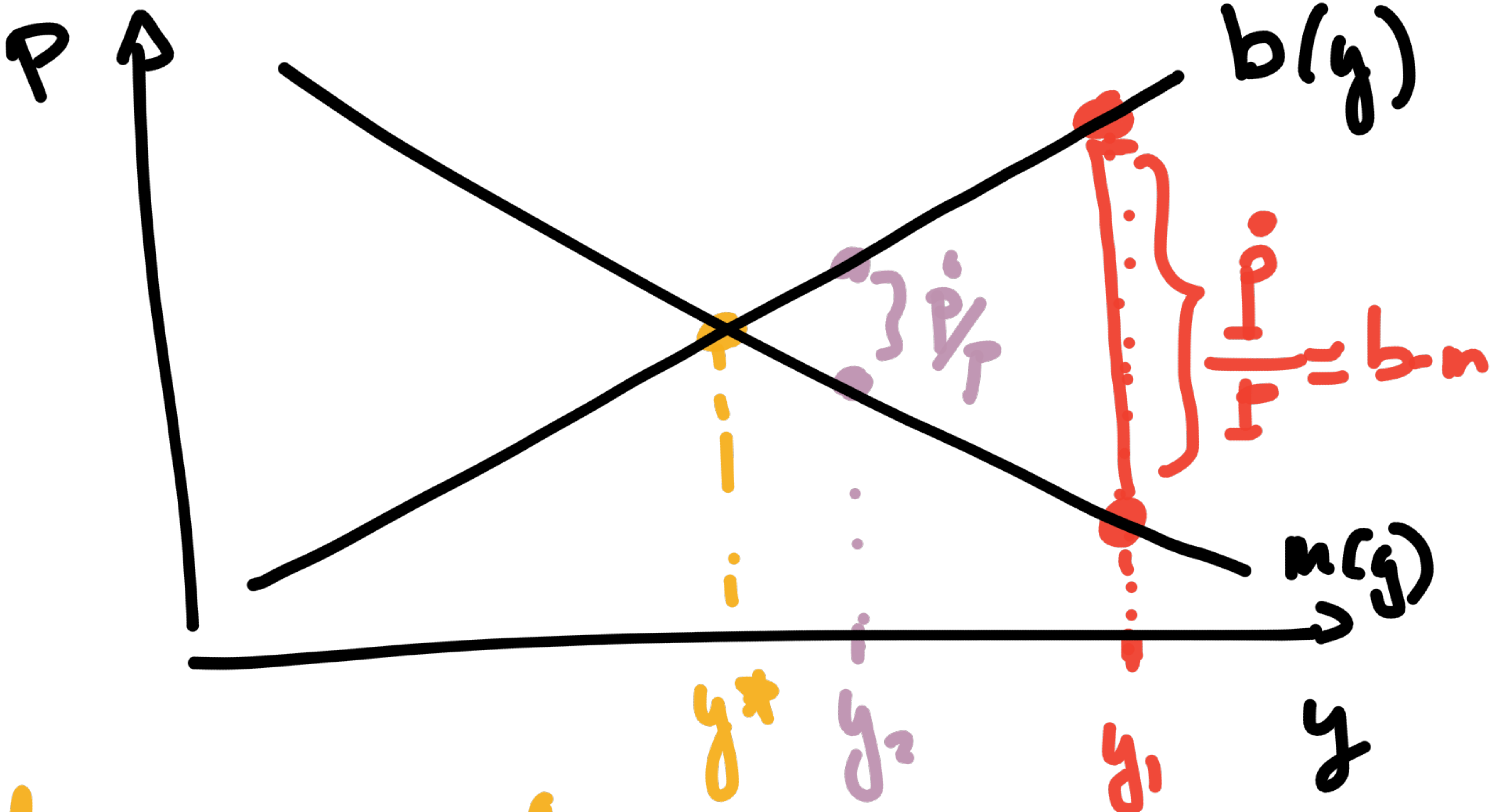
$$\dot{P} = bP - mP$$

$$\dot{P} = (b-m)P$$

$$\underbrace{\frac{\dot{P}}{P}} = b-m$$

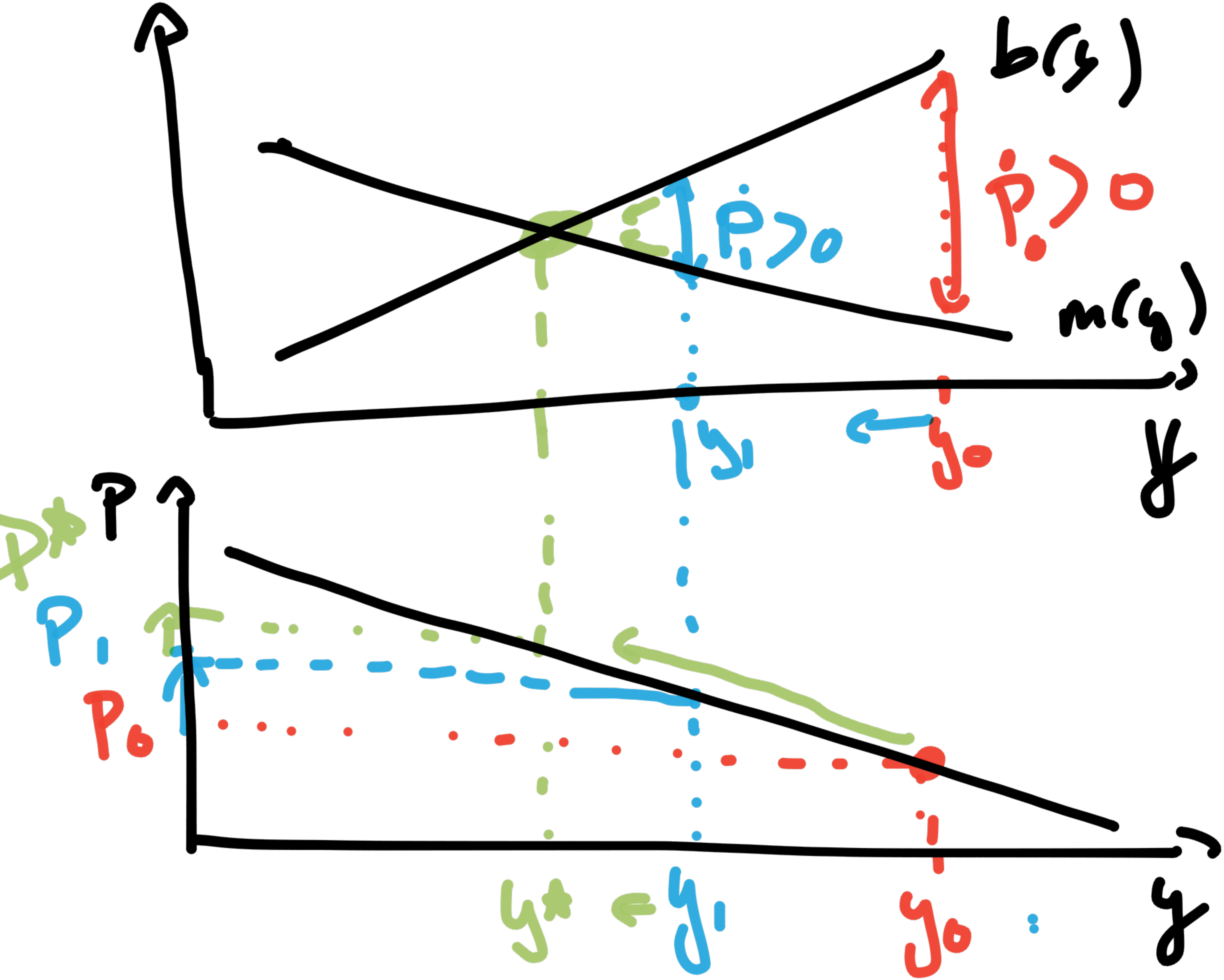
→ growth rate of populat.

<u>SS</u> : $\dot{P}=0$

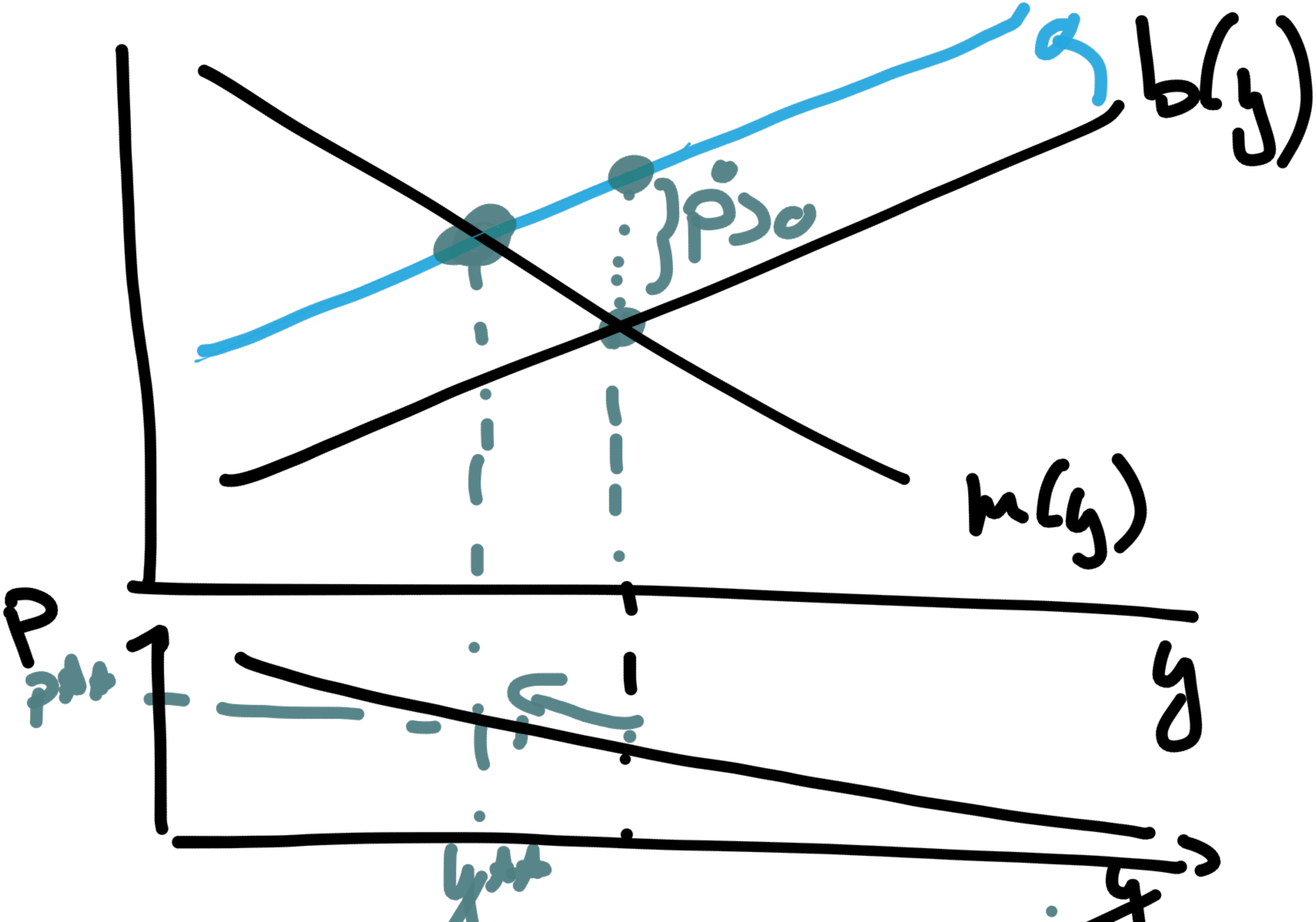


if $y = y^*$, $\dot{P} = 0$ because $m(y^*) = b(y^*)$

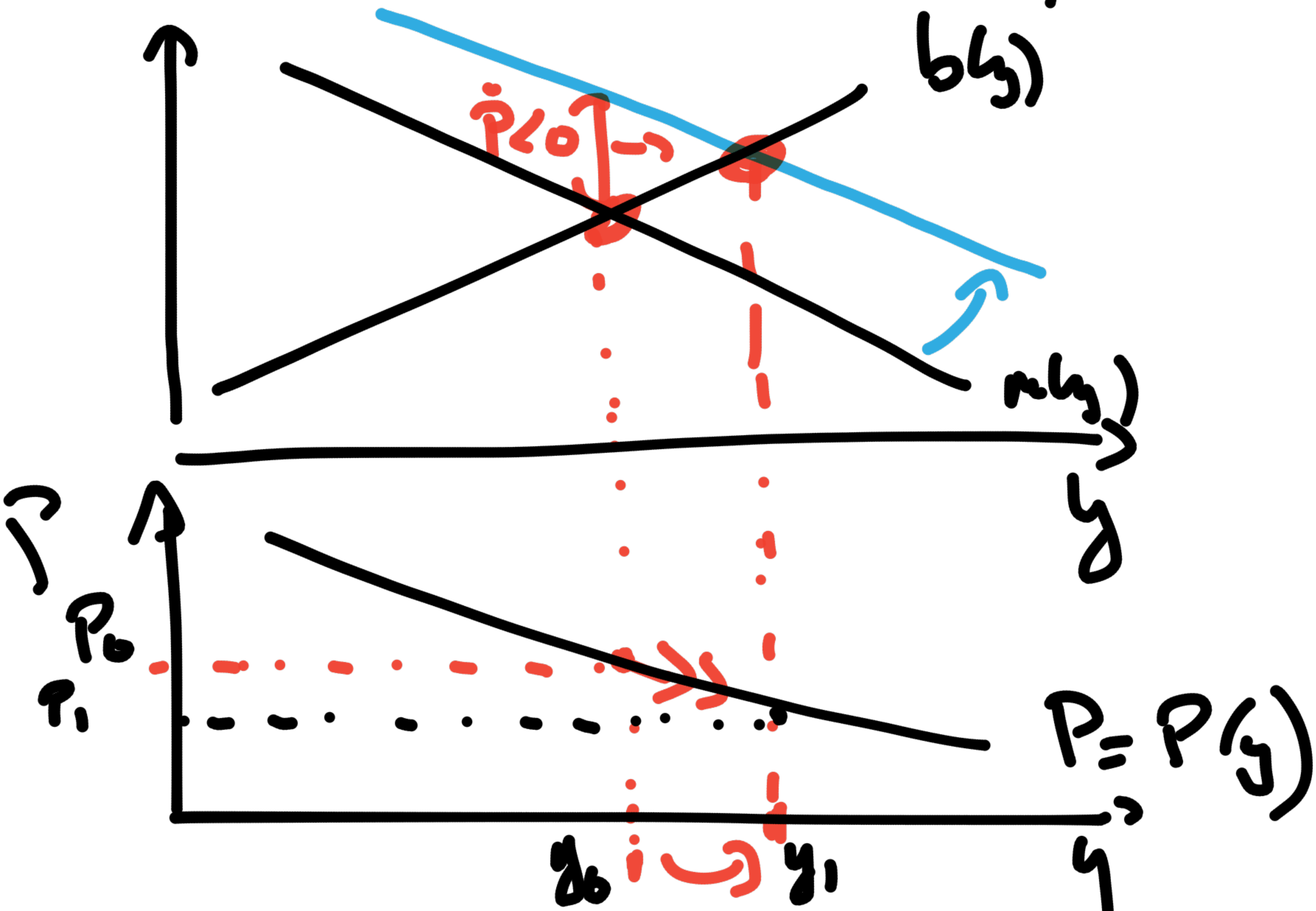
↳ Ss

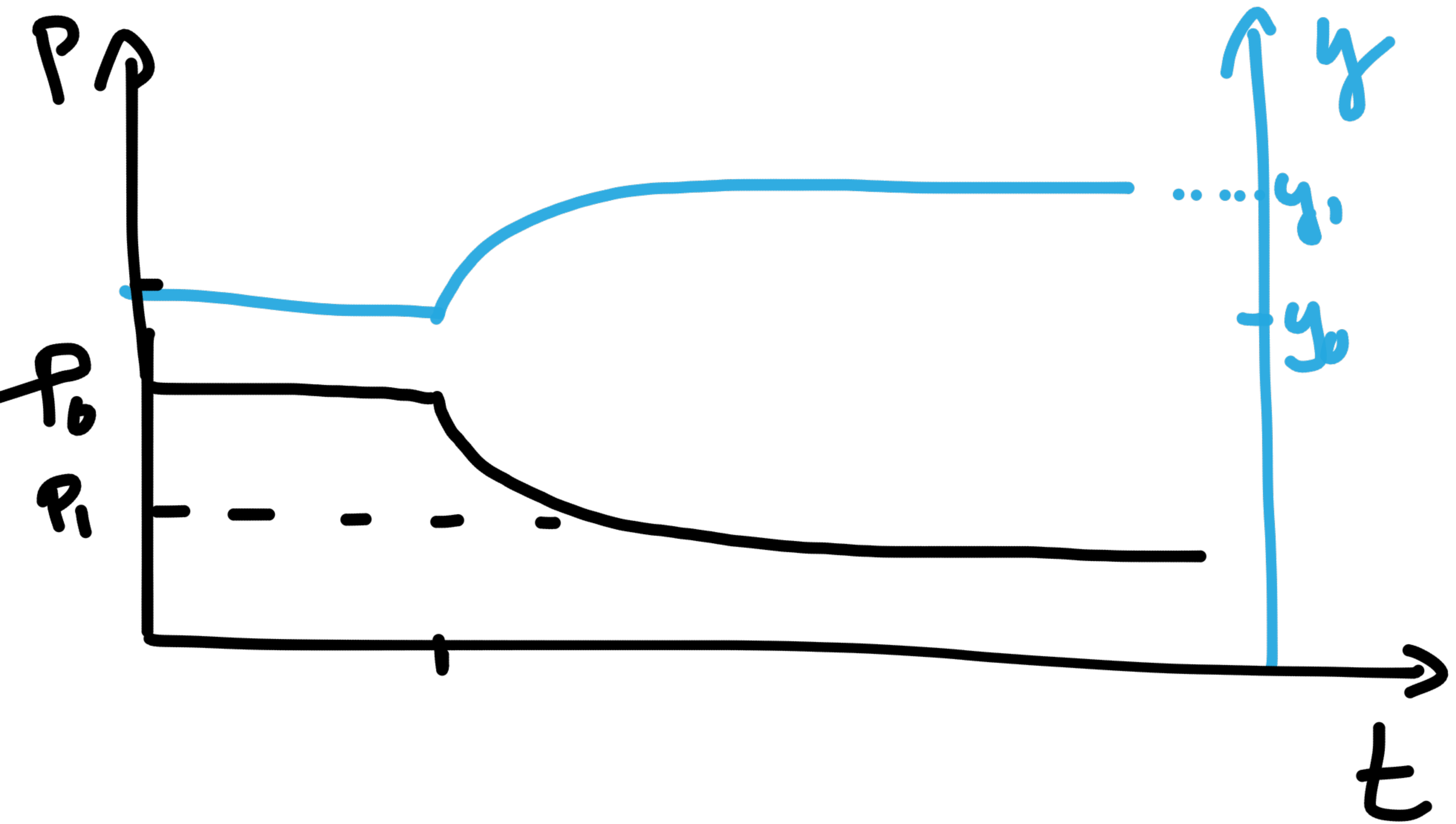

b(y)
m(y)
ṗ > 0
ṗ₁ > 0
y₁
y₀
y
p
p*
p₁
p₀
y* ⇐ y₁
y₀
y

What if b(y) shifts upwards



Black Plague : m(y) shifts




P
y
y_1
y_0
P_0
P_1
t